



College of Engineering

MD-024

Computer Vision-Based Settling Rate
Analysis for Water Treatment
Project Proposal

Prepared for

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Executive Summary

Access to clean water is essential for all forms of life. That being said, having an accurate way to determine what's in your water is a very important issue to solve. The treatment process involves dosing or adding coagulants that are used to eliminate suspended solids and particles found in water. We then have to determine the correct dosage and coagulant selection and allow operators to perform jar tests, which simulate the coagulation process in a lab setting. The issue is that traditional jar testing relies on visual observation to judge settling behavior, making it slow, subjective, and prone to inconsistency.

This project aims to address these limitations by developing a portable, computer vision-based system that uses computer vision and machine learning to measure surface settling rates and predict turbidity reduction. The goal is to replace manual, visual estimates from jar tests with consistent, data-driven measurements that improve accuracy, reliability, and repeatability.

The prototype integrates a camera, a controlled lighting system, and custom software designed to analyze footage and make the jar testing process seamless. It tracks the movement of the interface and particles, extracting features such as texture changes, settling curves, and turbidity predictions in real time. Results are displayed live, allowing operators to compare jars side by side and make faster, more informed decisions.

The system will be validated according to quality standards that are EPA Method 180.1 and ISO 7027. The standard methods used for turbidity measurement will include Method 210B and ASTM D2035. The design will emphasize portability, quick setup, and compatibility with data integration in Excel.

The key deliverables include the imaging system, custom software, and a labeled dataset in Excel, along with standard operating procedures and a verification report documenting the system's accuracy and reliability. Success will be measured by improved consistency compared to manual jar testing and the system's ability to produce clear, repeatable results. This project aims to bring automation and objectivity to a process that has remained manual for decades, ultimately improving efficiency and usability.

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Section A. Problem Statement

Water treatment relies on the effective removal of suspended solids and other contaminants to ensure water quality and safety. This process is not only important for public health, but also for supporting industrial operations and environmental sustainability^[1]. ChemTreat is a leader in the water treatment industry, designing industrial water treatment solutions for a variety of customers in industries such as healthcare, chemical processing, and food and beverage^[2]. These customers rely on water treatment solutions that enhance system efficiency, reduce operational costs, and ensure compliance with regulatory and quality standards.

To achieve these outcomes, it is crucial to remove contaminants early in the process through effective treatment methods. The first critical step in water treatment is the addition of coagulants to bind suspended particles in raw water to form larger clumps, accelerating their settling and removal^[1]. A variety of coagulants are able to accommodate the unique characteristics of different raw water sources and desired treatment outcomes. Because of this, tests must be conducted to identify the best coagulant to suit the unique needs for each water treatment process. This ensures the selected coagulant performs effectively to optimize treatment efficiency and water quality^[3].

Jar testing is used to evaluate the effectiveness of a coagulant in facilitating solid-liquid separation by mimicking a full scale water process in a smaller environment. It does this by varying the type, dosage, and sequence of treatment chemicals across multiple water samples which are then subjected to a treatment process involving coagulation, flocculation, and filtration in order to assess the best coagulant to their system^[3,4]. The current limitation with this technology is that it relies on manual testing procedures and subjective observation, which introduces variability between operators and increases the likelihood of human error. Factors such as individual judgement and procedural inconsistencies can all affect the accuracy and repeatability of results. Additionally, traditional jar testing is typically confined to laboratory settings which limits its responsiveness to changing conditions in the field.

There is a growing need for this technology to evolve into a more automated and portable solution that is capable of being performed on-site^[4]. Developing a more adaptable and efficient testing method would improve the accuracy and speed of water treatment decisions across a variety of operating conditions. The need for clean water affects everyone, and in environments where clean water is limited, making swift decisions is essential to ensuring reliable access to clean and safe water. To meet this need, our team is aiming to utilize vision-based computer technology through machine learning and computer vision to create a portable apparatus that can quantifiably measure solid settling rates.

Previously, another student team designed a prototype around this vision-based learning model. It showed much success in proving concepts, allowing short-term predictions of settling rates. However, there were still some limitations this project seeks to improve upon. For one, the prototype still requires a good amount of setup and does not quite meet the mobility requirements desired by ChemTreat. Secondly, the prototype is only able to predict one water

sample at a time, which does not reflect the current industry standard and needs to have comparative testing.

There was a similar study involving machine learning and water treatment conducted in Japan that used convolutional neural networks (CNN) to optimize dosage of polyaluminum chloride, a coagulant. The experiment used a Sony RX1R digital camera to record 30 jar tests for 22 minutes each. Over a million pictures were collected and processed using an NVIDIA Geforce gtx 1080 GPU. While the tests were catered to Japanese water standards, the study shows that machine learning is applicable and beneficial to water treatment^[5].

Other attempts at using computer vision-based learning to predict settling rates have also been used, with similar issues as this project's predecessors. A prototype machine and software, SWAMVA, made lots of headway in using machine learning and vision-based computer technology to accurately and quickly predict solid settling rates, greatly reducing prediction times.^[6] SWARMVA, however, still struggles with mobility, accessibility, and small sample sizing.

These previous researchers prove the viability of using technology to aid in water treatment assessment instead of relying on subjective measures from field testers. Furthermore, making the settling rate predictions accessible to someone in the field rather than waiting for turbidity tests from a lab will greatly increase timely client assessments. While ChemTreat is the main client for this project, these solutions will serve a diverse range of stakeholders including industrial water treatment companies and their stakeholders, regulatory agencies, environmental groups, and communities that depend on safe, quality water. By creating a portable method for quantifying solid settling times in water treatment, this technology will support treatment operations and improve adaptability of water systems.

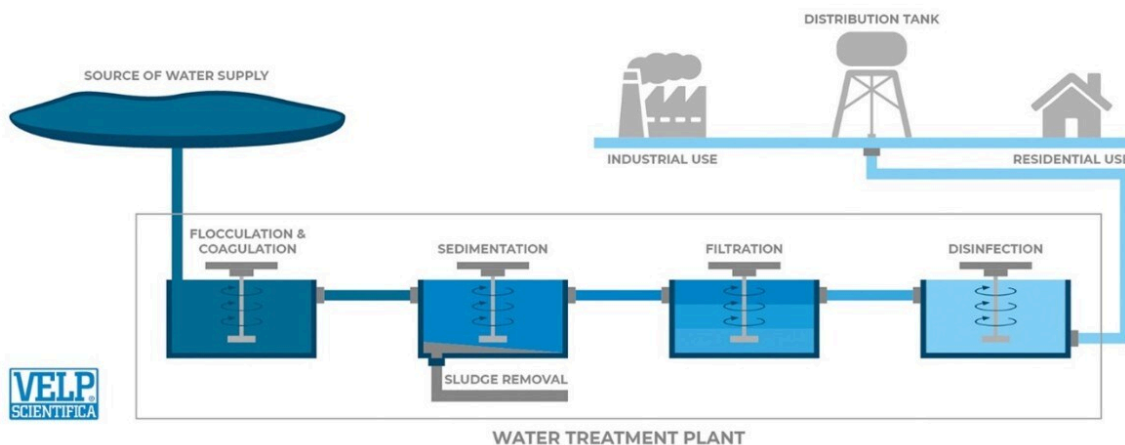


Figure 1. From Fisher Scientific Lab Reporter (2023, Issue 3) [1].

Section B. Engineering Design Requirements

B.1 Project Goals (i.e. Client Needs)

From the client's perspective, the purpose of this project is to improve how settling performance is assessed during jar tests and pilot runs. The client values results that are objective, fast, and easy to compare across chemicals and doses. The goals below are written as outcomes the client needs, not descriptions of the design.

- We want to deliver an object and repeatable standardized settling measurements to then replace the manual reading of the jar tests.
- We want to make the processes faster and the decision making and the reading of the jar test more consistent.
- We want to be able to provide clear and realtime results that are all met within the safety standards and be within the budget.

B.2 Design Objectives

The overall objective of this product's design is to predict the effectiveness of different treatments for settling suspended solids in water. This design seeks to improve upon the established jar test by creating a setup that is more portable and accessible for researchers who travel to client sites to provide estimations for effective treatments. The system will use machine learning and computer vision tools to build and train a model that can accurately predict settling behavior. Since the settling time of suspended solids typically ranges from 5 to 20 minutes depending on the jar sample, the goal is to predict the final settling time within 3 to 5 minutes, saving up to 15 minutes compared to traditional methods. The design aims to deliver an objective, repeatable, and standardized approach to settling measurements, replacing the subjective visual estimates currently used throughout the industry. This test will allow for more data based decisions, ensuring consistency and comparability across operators, chemicals, doses, and various raw water sources. The system is intended to integrate seamlessly into existing lab workflows while adhering to all safety standards, budget constraints, and project limitations.

B.3 Design Specifications and Constraints

Below is a list of design specifications and constraints that the project will follow:

- Video recording requirements
 - Video must be recorded with a resolution of at least 1080p
 - Video will be recorded with at least 15 fps
 - Video should be recorded using a smartphone
- Machine learning requirements
 - Device must have at least 8 GB of RAM

- Device must be at least 256 GB of storage
- Program must use publicly available machine learning and computer vision libraries
- Program must be able to analyze between two to four jars side by side
- The trained model must return a prediction of final settling time within 3-5 minutes of the jar test

B.4 Codes and Standards

Below is the list of specific codes and standards that apply to this project, with a brief note on how each one relates to the design, testing, and operation of the prototype. Standards guide the technical methods and quality expectations. Codes are legal or regulatory requirements that must be met.

1.EPA Method 180.1 – References the method for turbidity used for the ground truth comparison of predicted NTU values in low ranges.

2.ISO 7027-1:2016 – This is the international turbidity standard and the turbometric procedures used to validate low and high turbidity waters.

3.ASTM D2035 –This is the standard practice for coagulation for jar tests and defines the variables dosing steps and the evaluation criteria.

4.Standard Methods 2130 B – This is the lab procedure for turbidity and this is to ensure the collaboration to align with the common lab practices.

5.IEC 61010-1 is the safety requirements for electrical equipment used in the laboratory to measure and control the lights and the camera and finally the rig itself.

Section C. Scope of Work

C.1 Deliverables

The final prototype of this project will have two major components to it: the physical testing rig and the coding and analysis of data collection. For the physical testing rig, it will contain two to three rectangular jars side-by-side so that a video can be collected of all three jars simultaneously. This is to ensure we satisfy the comparative component desired from the project by ChemTreat as it better aligns with timeliness and current jar test designs. These jars should be around 250ml in size to ensure accurate reflection of the sampled water's characteristics and give enough suspended solids presents to be settled. Furthermore, the jars will have an agitator in some form to appropriately mix the water samples collected before settling rates are measured. Furthermore, the physical rig will require a camera, one that should ideally be supplied by the tester in the form of their personal or work smartphone's camera. This meets the mobility and accessibility requirements of our Client, ChemTreat. This implies the need for a built in or supplied camera stabilizer and holder to obtain more accurate data for the treatment analysis. Similarly, lighting will also be an important factor to control for visual-based data collection. Thus, lightstrips should be attached to the physical rig. This to better assess turbidity of the water to then correlate it to the settling rates of the suspended solids in the water sample. The physical rig hopes to best eliminate any disturbances that the video processing might have to work around while maintaining the overall goal of a mobile and accessible prototype.

The data analysis aspect of this project also has to keep the mobility and time frame constraints in mind. The video will be first gathered on a smartphone camera that meets the 1080p and 15 fps requirements. This video will then be sent to a computer more capable of machine learning program execution. This video should then be analyzed and able to return a comparative settling rates analysis within the first five minutes of filming. This will require an exportation of the video's data to a third party program that can then interpret the visual data as it relates the water sample's turbidity, learn expected behaviors, and make settling predictions off that. That returned data should entail a simple and easily digestible comparative analysis of the two to four jars present in the physical rig. That is to say, the coding should return expected settling rate times and effectiveness of that settling for each jar test and compare the jars to one another. This is to quickly gather which coagulant, flocculant or other treatment would be best suited for the sampled water in a timely manner and in a way that most field testers can understand.

Overall the prototype aims to give preliminary settling rate estimates of two to four water samples simultaneously and compare the results. The actual settling rate tests will only need to last some 5 minutes and predict the next 15 to 25 minutes of solid settling behavior in the sample water. After that 5 minute data collection is done, the video used to record should be sent for interpretation and return an easily comprehensible and comparative analysis.

C.2 Milestones

<u>Tasks</u>	<u>Timeframe</u>
Team Contract	8/27/2025 - 9/5/2025
Project Design Proposal	9/5/2025 - 10/10/2025
<i>Creation</i>	<i>9/5/2025 - 10/7/2025</i>
<i>Review</i>	<i>10/7/2025 - 10/10/2025</i>
Apparatus Development	10/7/2025 - Spring Semester
<i>First Prototype</i>	<i>10/7/2025 - 12/08/2025</i>
<i>Have purchased (all known at that point) necessary equipment for the apparatus</i>	<i>11/21/2025</i>
<i>Final Prototype</i>	<i>Spring Semester</i>
Model Development	10/31/2025 - Spring Semester
<i>Code Creation</i>	<i>10/31/2025 - Into Spring Semester</i>
<i>Code Testing</i>	<i>10/31/2025 - Into Spring Semester</i>
<i>Data Collection and Training</i>	<i>Spring Semester - After first prototype is settled on</i>
Fall Semester Deadlines	
<i>Project Proposal - Deadline</i>	<i>10/10/2025</i>
<i>Fall Design Poster - Deadline</i>	<i>11/14/2025</i>
<i>Preliminary Design Report - Deadline</i>	<i>12/8/2025</i>

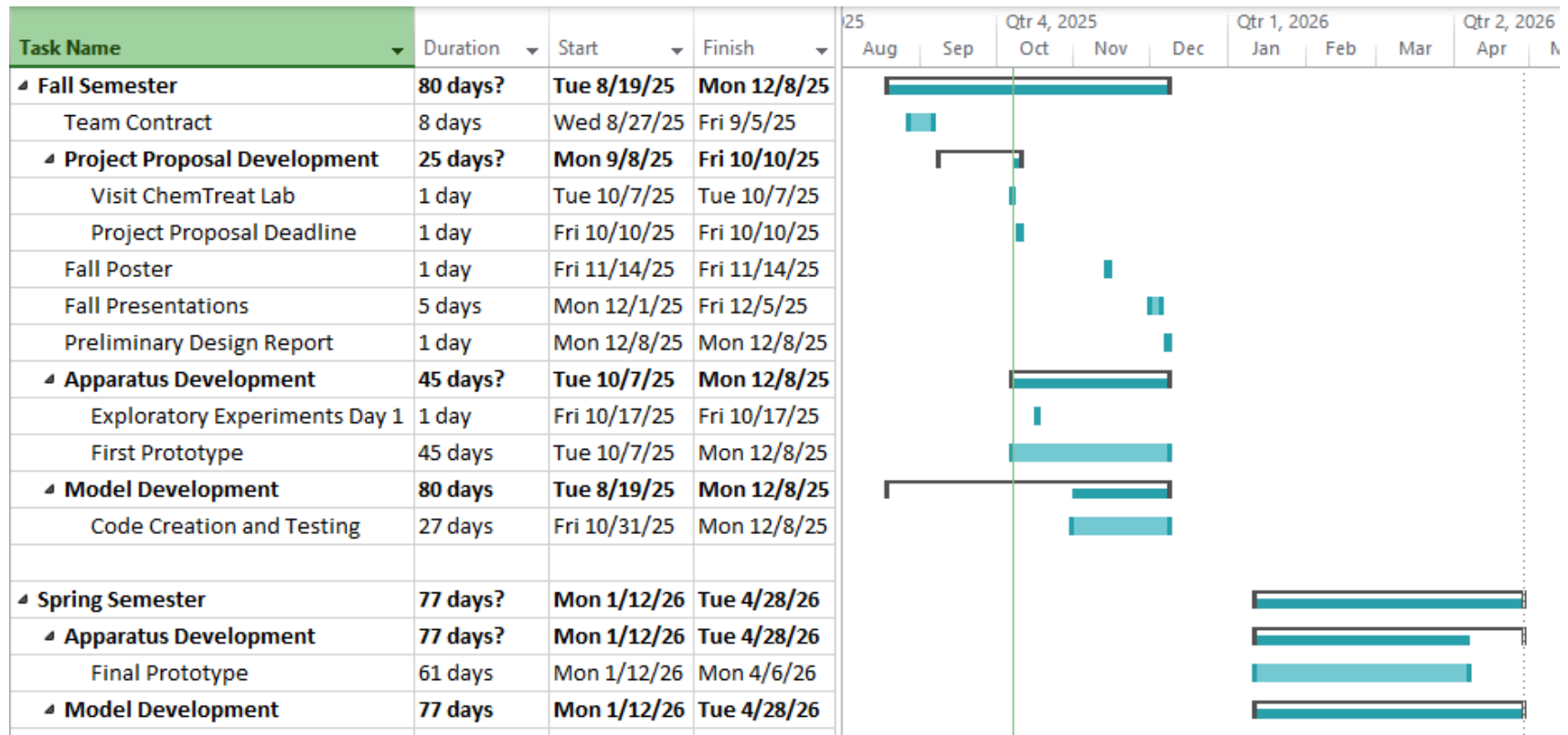
Section D. Future Work, Conclusions, and Recommendations

This project is aiming to use a computer vision-based, portable apparatus similar to jar tests that can quantifiably measure solid settling rates through machine learning to improve the accuracy and speed of treatment chemical selection in water treatment. The primary objectives are to obtain objective and repeatable settling measurements, enable testing in the field with easily understandable results, and create consistent comparisons across treatment chemicals, doses, and raw water sources. Our future plans are to evaluate the current technology such as jar tests and the current prototype from last year's capstone project. This evaluation will include a series of exploratory laboratory tests designed to identify their limitations and assess their performance. Following this assessment, we will develop our own prototype that incorporates our desired improvements based on our findings and the project goals. Throughout the design process, we will ensure our prototype is benchmarked against established industry standards to ensure technical relevance and expected performance. The goal is to create a portable benchtop rig compatible with iPhone technology that can support 2 to 4 jars. This system will enable quantifiable measurement of suspended solids and real-time turbidity characterization over time with live settling metrics, a comparative ranking system, and data export capabilities.

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<https://doi.org/10.1016/j.aca.2021.339411>

Appendix 1: Project Timeline



Appendix 2: Team Contract (i.e. Team Organization)

Step 1: Get to Know One Another. Gather Basic Information.

Task: This initial time together is important to form a strong team dynamic and get to know each other more as people outside of class time. Consider ways to develop positive working relationships with others, while remaining open and personal. Learn each other's strengths and discuss good/bad team experiences. This is also a good opportunity to start to better understand each other's communication and working styles.

<i>Team Member Name</i>	<i>Strengths each member bring to the group</i>	<i>Other Info</i>	<i>Contact Info</i>
Aubrey Carey	Lab Research Experience Coding Industry Experience (Process Controls)		careyae2@vcu.edu 703-853-2511
Rohan Sharma	Coding Great Questions! Debugging App Design		sharmar16@vcu.edu 571-831-1762
Virginia Anderson	Communication skills Chemistry Some Water treatment knowledge		andersonvc2@vcu.edu (757)-947-4942
Ronald Menendez	Coding Problem solving skills		menendezrd@vcu.edu (804)-432-6759

<i>Other Stakeholders</i>	<i>Notes</i>	<i>Contact Info</i>
Dr. McGill Dr. Thomson-McInnes	<i>Faculty Advisors</i>	mcgillc2@vcu.edu btmcinnes@vcu.edu
Byamungu Ronald Praise Adebisi	<i>Project Management Advisors</i>	ronaldb@vcu.edu adebisip@vcu.edu
Vlad Djukanovic Will Henderson	<i>Sponsors</i>	vladimird@chemtreat.com williamh@chemtreat.com

Step 2: Team Culture. Clarify the Group's Purpose and Culture Goals.

Task: Discuss how each team member wants to be treated to encourage them to make valuable contributions to the group and how each team member would like to feel recognized for their efforts. Discuss how the team will foster an environment where each team member feels they are accountable for their actions and the way they contribute to the project. These are your Culture Goals (left column). How do the students demonstrate these culture goals? These are your Actions (middle column). Finally, how do students deviate from the team's culture goals? What are ways that other team members can notice when that culture goal is no longer being honored in team dynamics? These are your Warning Signs (right column).

Resources: More information and an example Team Culture can be found in the Biodesign Student Guide "Intentional Teamwork" page ([webpage](#) | [PDF](#))

<i>Culture Goals</i>	<i>Actions</i>	<i>Warning Signs</i>
Constructive Feedback	● Set aside time in team meetings to review one another's work and offer suggestions	Editing other's work without consultation
Be proactive	● Setting personal goals of what needs to be done by next group meeting ● Doing tasks in parallel and working together to ensure deadlines are met ● Not leaving things till the last minute	Missing deadlines Holding up the team Not showing up to meetings
Clear Communication	● Communicate workload via individual and group action items ● Organizing files ● Ensuring teammates understand the work you're doing	Not responding in text groupchat/discord

Step 3: Time Commitments, Meeting Structure, and Communication

Task: Discuss the anticipated time commitments for the group project. Consider the following questions (don't answer these questions in the box below):

- What are reasonable time commitments for everyone to invest in this project?
- What other activities and commitments do group members have in their lives?
- How will we communicate with each other?
- When will we meet as a team? Where will we meet? How Often?
- Who will run the meetings? Will there be an assigned team leader or scribe? Does that position rotate or will same person take on that role for the duration of the project?

Required: How often you will meet with your faculty advisor, where you will meet, and how the meetings will be conducted. Who arranges these meetings?

See examples below.

<i>Meeting Participants</i>	<i>Frequency Dates and Times / Locations</i>	<i>Meeting Goals Responsible Party</i>
Students Only	Once a week, on Tuesdays via Online calling and video	Update group on day-to-day challenges and accomplishments Assign or delegate work needed for project to progress Actively work on project
Students + Faculty advisor	Once a week most weeks, on Fridays at 11 am in W101	Update faculty advisor and get answers to our questions Set up/update calendar with hard deadlines
Project Sponsor	TBD Meet in person or by zoom	Update project sponsor and make sure we are on the right track

Step 4: Determine Individual Roles and Responsibilities

Task: As part of the Capstone Team experience, each member will take on a leadership role, *in addition to* contributing to the overall weekly action items for the project. Some common leadership roles for Capstone projects are listed below. Other roles may be assigned with approval of your faculty advisor as deemed fit for the project. For the entirety of the project, you should communicate progress to your advisor specifically with regard to your role.

- **Before meeting with your team**, take some time to ask yourself: what is my “natural” role in this group (strengths)? How can I use this experience to help me grow and develop more?
- **As a group**, discuss the various tasks needed for the project and role preferences. Then assign roles in the table on the next page. Try to create a team dynamic that is fair and equitable, while promoting the strengths of each member.

Communication Leaders

Suggested: Assign a team member to be the primary contact for the client/sponsor. This person will schedule meetings, send updates, and ensure deliverables are met.

Suggested: Assign a team member to be the primary contact for faculty advisor. This person will schedule meetings, send updates, and ensure deliverables are met.

Common Leadership Roles for Capstone

1. **Project Manager:** Manages all tasks; develops overall schedule for project; writes agendas and runs meetings; reviews and monitors individual action items; creates an environment where team members are respected, take risks and feel safe expressing their ideas.
Required: On Edusourced, under the Team tab, make sure that this student is assigned the Project Manager role. This is required so that Capstone program staff can easily identify a single contact person, especially for items like Purchasing and Receiving project supplies.
2. **Logistics Manager:** coordinates all internal and external interactions; lead in establishing contact within and outside of organization, following up on communication of commitments, obtaining information for the team; documents meeting minutes; manages facility and resource usage.
3. **Financial Manager:** researches/benchmarks technical purchases and acquisitions; conducts pricing analysis and budget justifications on proposed purchases; carries out team purchase requests; monitors team budget.
4. **Systems Engineer:** analyzes Client initial design specification and leads establishment of product specifications; monitors, coordinates and manages integration of sub-systems in the prototype; develops and recommends system architecture and manages product interfaces.
5. **Test Engineer:** oversees experimental design, test plan, procedures and data analysis; acquires data acquisition equipment and any necessary software; establishes test protocols and schedules; oversees statistical analysis of results; leads presentation of experimental finding and resulting recommendations.
6. **Manufacturing Engineer:** coordinates all fabrication required to meet final prototype requirements; oversees that all engineering drawings meet the requirements of machine shop or vendor; reviews designs to ensure design for manufacturing; determines realistic timing for fabrication and quality; develops schedule for all manufacturing.

<i>Team Member</i>	<i>Role(s)</i>	<i>Responsibilities</i>
Aubrey Carey	Project Manager	• Develop overall schedule for project • Runs meetings and makes overview slides • Reviews and monitors individual action items
Virginia Anderson	Lead Experimenter (ChemE); Sponsor Contact	• Lead person for design of experiments • Manages lab notebook and data collection • Establish contact with ChemTreat • Document meeting minutes
Rohan Sharma	Lead Developer (CS); Faculty Advisor Contact	• Manages the GitHub • Schedule meetings, send updates, and ensure deliverables are met for faculty advisors • Mentor team members by providing guidance and support to help team members improve their skills
Ronald Menendez	Financial Manager	• Research technical purchases and acquisitions • Conduct pricing analysis and budget justifications on proposed purchases • Carries out team purchase requests and monitors the budget

Step 5: Agree to the above team contract

Team Member: Aubrey Carey

*Signature: **Aubrey Carey***

Team Member: Virginia Anderson

*Signature: **Virginia Anderson***

Team Member: Rohan Sharma

Signature: Rohans

Team Member: Ronald Menendez

Signature: Ronald Menendez